

Project Goal | *UK based PC that will compete with the new IBM PC.*

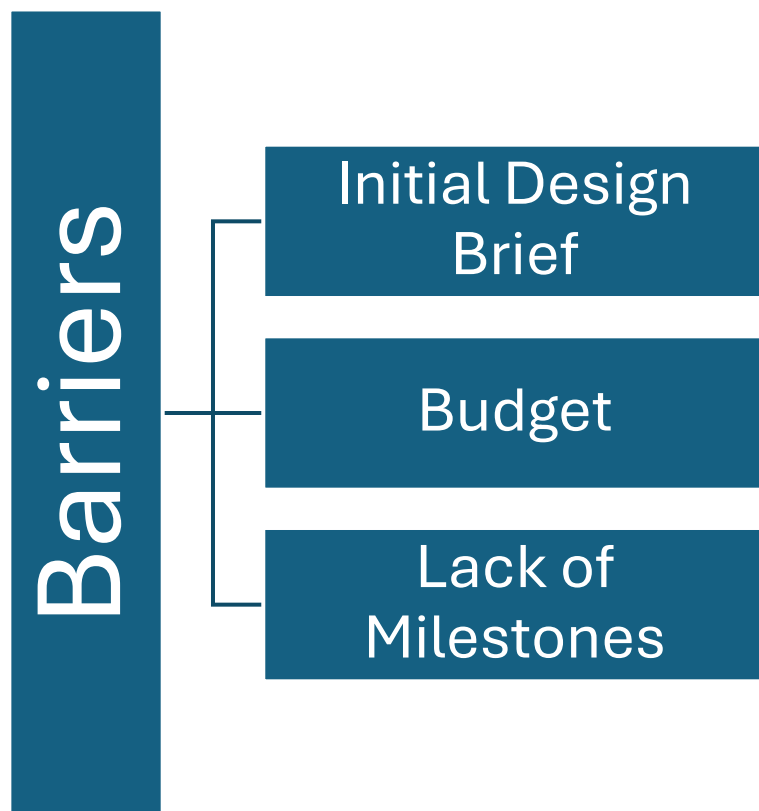
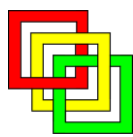
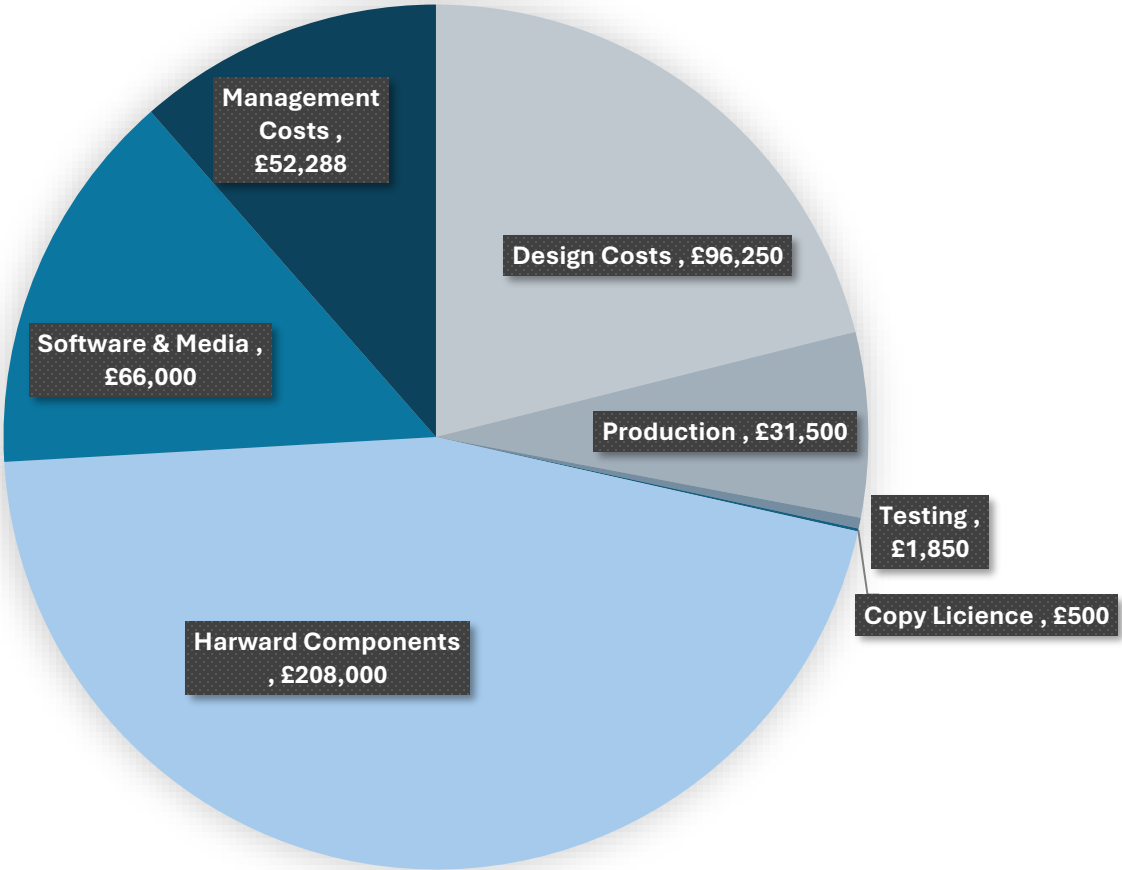


Figure 1 – Barriers to Success

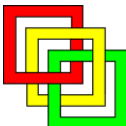
	The initial design brief was based on an informal conversation between Will and Colin.
	The budget was unrealistic. The IBM PC retails at £1000 (Edwards, 2021), the cost for the Symputer is just £250.
	The lack of milestones and review meant a lack of agility.



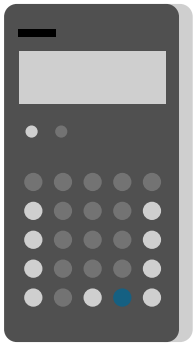
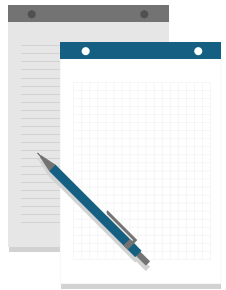
Synputer Mark I



 	Contingency £43,612 Contingency is less than 10%
	2000 units have been produced Synful Computing Contingency £21 per unit EDC £150 margin per unit £300,000
	The solution will be to evolve and develop the Synputer Mark II



EDC | *Updated Requirements (Nov 1983)*



Industry standard operating system

- Syn OS provided / Unix at additional cost

At least a 68000 CPU – preferably upgradable

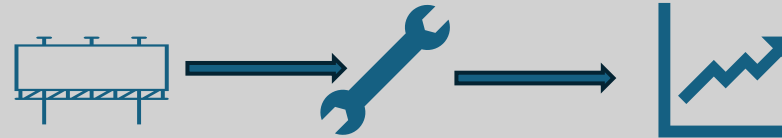
- Provided in Synputer Mark I

SCSI expansion capability

2 serial ports that support RS 422/485 standard

Support a GUI system and mouse

- Upgrade Option: Pro Expansion Card



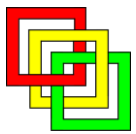
External keyboard/ connector

- Included as standard in the Synputer Mark II

At least 512KB of RAM



At least 1 industry standard drive with removable media



Synputer Mark II | Tomorrow's Technology Today

Updated Requirements | *Solutions (1)*

Industry standard operating system

Supply Unix and Copy Licence

- Syn OS can be considered an industry standard
- Unix can be supplied with Synputer Mark I
- Additional Cost £10,599. Assume FX £1:\$1
- EDC can realise *their* priority 1



Solution | **£10,500**

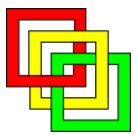
External keyboard/ connector

Include in Synputer Mark II

- Upgrade Case @ £10 per Unit
- Architecture Redesign @ £6,250
- Upgrade Keyboard @ £2.50 per Unit
- Redesign G1 IOP Chip @ £2500



Solution | **£33,750**



Updated Requirements | *Solutions (2)*

At least 512KB of RAM

Include in Synputer Mark II / Upgrade

- Mark I delivers functionality using 128KB
- No need to delay launch of Mark I
- Upgrade to 512KB with Mark II @ £4 per Unit
- Redesign G2 RAM Chip @ £2500
- Additional RAM can be provided via the Pro Expansion Card upgrade



Solution | **£10,500**

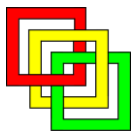
1 industry standard drive with removable media

Include in Synputer Mark II

- Mixed Storage: Cartridge and Floppy Disc
- Upgrade Storage @ £7.50 per Unit
- Architecture Redesign @ £2500



Solution | **£15,000**



Updated Requirements | *Solutions (3)*

SCSI expansion capability

Include in Synputer Mark II / Upgrade

- SCSI delivered through the Pro-Expansion Card
- Cannot be supplied as an upgrade to the Synputer Mark I
- Pro-Expansion Card @ £15 per unit
- SCSI interface & terminator @ £5 per unit
- Redesign G2 RAM Chip @ £2500



Solution | **£42,500**

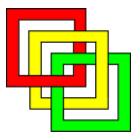
At least a 68000 CPU, preferably upgradable

Development for Synputer Mark II

- 68000 CPU included in Mark I
- Upgrade to socketed board for Mark II @ £10 per Unit



Solution | **£20,000**



Updated Requirements | *Solutions (4)*

2 serial ports that support RS 422/ 485

Include in Synputer Mark II

- Upgrade to Multiplex SC150 @ £3 per unit
- Board redesign @ £1250
- Supports external keyboard
- Supports external mouse



Solution | **£7,250**

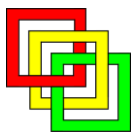
A board that is ready to support a GUI system and mouse if required by the user

Include in Synputer Mark II

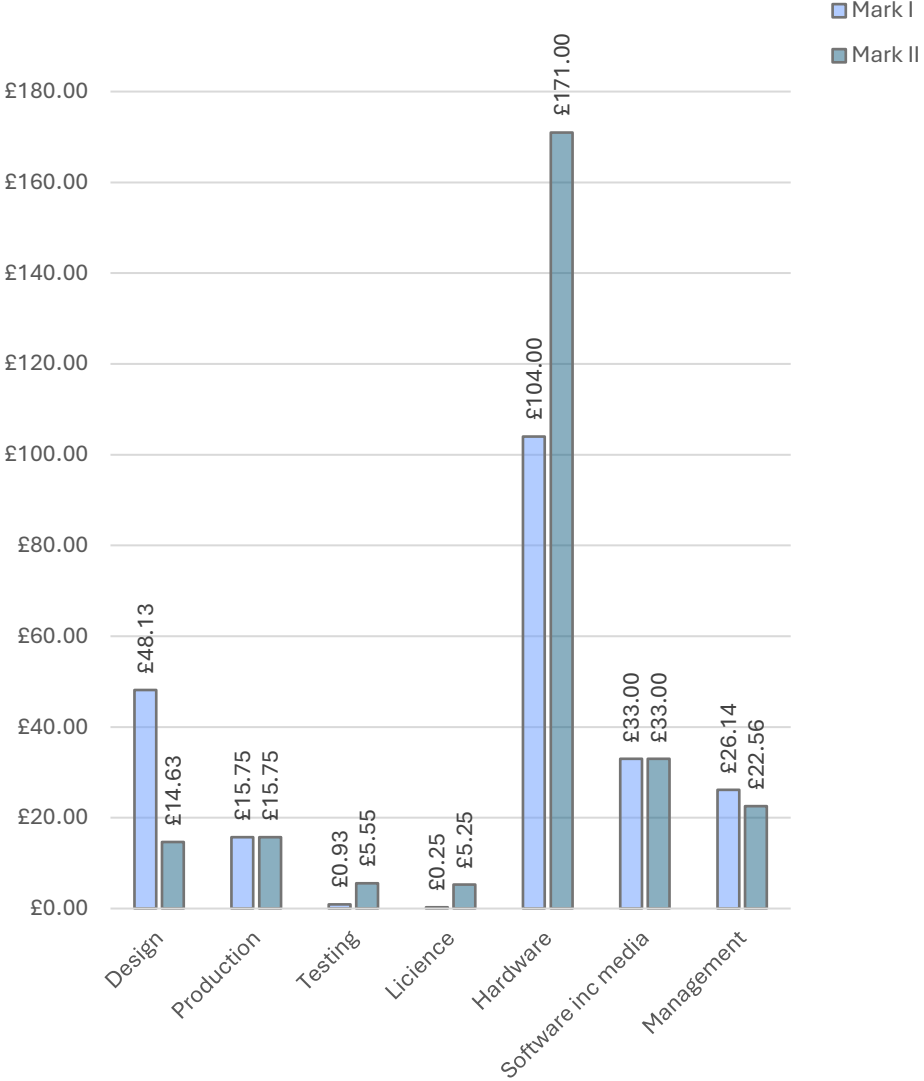
- The Mark II will pro upgrade to 64KB ROM @ £4 per Unit
- The Mark II will contain the hardware architecture to support a range of GUI
- No need to commit to a specific GUI now



Solution | **£8,000**



Synputer Cost Profile



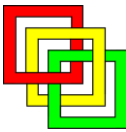
100% of EDC Revised Requirements Delivered



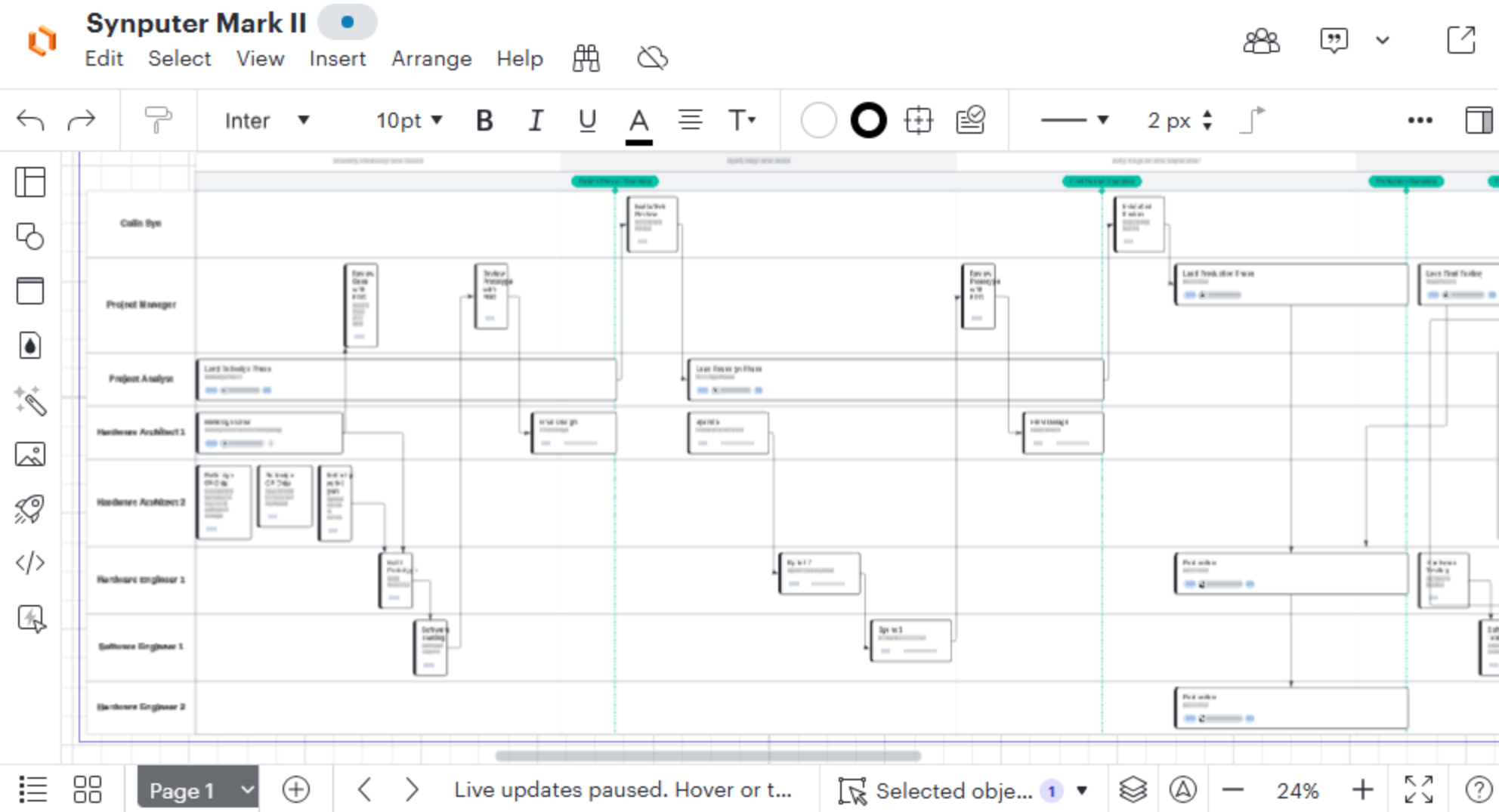
2000 Synputer Mark II to be produced at an agreed unit cost of just £275 (10% uplift)



UK based PC that will compete with the new IBM PC

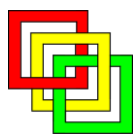


Synputer Mark II | *Agile Development*



Sprints

1. Case Design
2. Prototype
3. Final Design
4. Executive Review 1
5. Prototype 2
6. Final Design Amendments
7. Executive Review 2
8. Production
9. Hardware Testing
10. Software Testing
11. Executive Review 3
12. Launch Synputer Mark II



Synputer Mark II

*Large Print Copy In Handout Pack

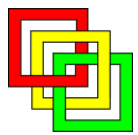
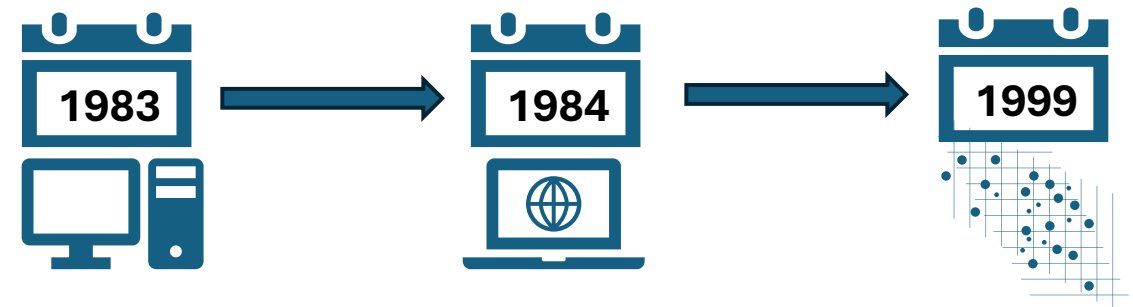
Synputer Mark II Cost per Unit | £275



Next Steps | *EDC-Synful Partnership*



	2000 Synputer Mark I Discounted Unit Price - £230 Launch December 1983
	2000 Synputer Mark II EDC Patnership Unit Price - £275 Launch December 1984



References

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